

ANTERVIEW

AI Voice Interview Platform

Project Sign-Off Document

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Classification	Confidential

1. Document Control

1.1 Revision History

Version	Date	Author	Description
1.0	26 June 2026	Intants Engineering	Initial issue for stakeholder sign-off

1.2 Distribution List

Recipient	Role / Interest
HR Team / HR Head	Primary users; owners of hiring workflow
Project Sponsor	Commercial approval and budget
Engineering Lead	Solution design and delivery
Compliance Officer	DPDP Act 2023 and data residency

2. Purpose of This Document

This **Project Sign-Off Document** defines the scope, solution design, functional flows, and acceptance criteria for the **Anterview AI Voice Interview Platform**. It gives HR and project stakeholders a single, shared reference for what will be delivered and how it works.

Signing this document records formal agreement that the **scope, approach, and acceptance criteria** described here are understood and approved, and authorises the project to proceed to delivery / demonstration.

3. Executive Summary

Anterview is a voice-first AI interview platform. A candidate logs in, picks a job role, and holds a natural spoken conversation with a realistic 3D avatar interviewer in their preferred language (**English, Hindi, Telugu** on Day-1, expanding to 22 Indian languages). The avatar speaks with lip-sync, listens, asks intelligent follow-ups, and produces a **structured scorecard** (on-screen and PDF) at the end of a ~10-minute session.

Beyond the candidate experience, Anterview provides a **multi-tenant HR suite**: bulk resume upload with AI ATS scoring, an exam module with three authoring methods (manual, AI-generated, Excel import), secure magic-link assessments and interviews, and an analytics dashboard. The platform is built as **four microservices** with an env-swappable, two-tier stack so the same code runs on a demo cloud today and on **AWS Mumbai** (India data residency) for production and government L1 bids, with a per-session variable cost target of **<= Rs. 12**.

4. Project Overview & Vision

Anterview replaces slow, inconsistent, and hard-to-scale first-round interviews with a fair, fast, voice-first AI experience available in the language the candidate actually thinks in. Every candidate receives a transparent, competency-level scorecard, and every recruiter receives structured, bias-checked results at population scale.

Target markets (priority order):

- Private engineering colleges & skill universities - fastest path to revenue.
- Corporate L&D and recruitment teams - mid-term recurring revenue.

- Government skilling bodies (APSSDC, NSDC, state SDCs) - long-term, high-value contracts via L1 bidding.

5. Objectives & Success Criteria

Objective	Target
Real-time conversation latency	p95 turn latency < 2 seconds
Availability	99.5% uptime (Multi-AZ design)
Scale	Up to 20 lakh (2 million) users
Session length	~10 minutes per interview
Language coverage	EN / HI / TE on Day-1; 22 Indian languages on roadmap
Unit economics	Variable cost <= Rs. 12 per session (target Rs. 10)
Compliance	DPDP Act 2023; India data residency (Mumbai) for production

6. Scope

6.1 In Scope

- Candidate AI voice interview with a 3D avatar (6 avatars: 3 male, 3 female).
- Real-time voice pipeline (speech-to-text, AI brain, text-to-speech, avatar lip-sync).
- AI scoring engine and structured scorecard (on-screen + PDF).
- Multi-tenant HR suite: bulk resume upload + AI ATS scoring and ranking.
- Exam module with three authoring methods: manual (4 options), AI-generated (Gemini), and Excel/CSV import.
- Secure magic-link assessment and interview invitations.
- Three-tier admin hierarchy + platform analytics dashboard.
- Pluggable authentication, SSO bridge (Naipunyam), and DPDP consent capture.
- Day-1 languages: English, Hindi, Telugu.

6.2 Out of Scope (this phase)

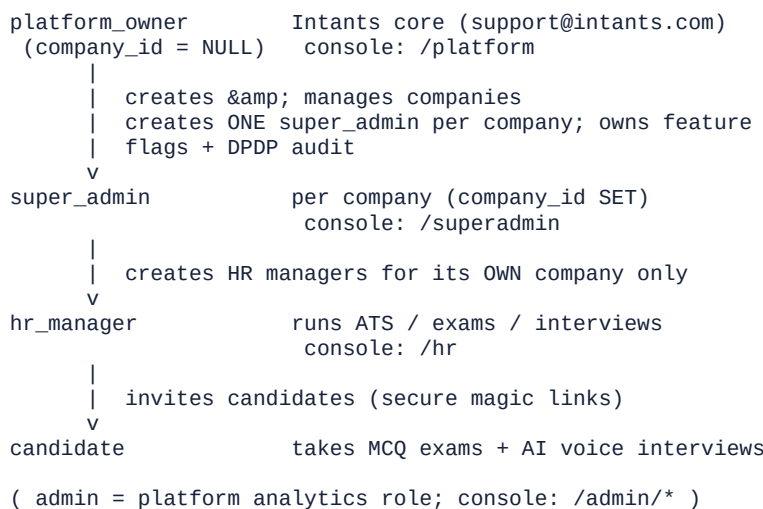
- Full 22-language coverage beyond EN/HI/TE (planned on the roadmap).
- In-house custom 3D avatar rendering pipeline (Tier-2 production option).
- Native iOS/Android apps (the responsive PWA covers mobile devices).
- Offline / no-internet interview mode.
- Third-party ATS/HRMS deep integrations beyond CSV export (future phase).

7. User Roles & Access Hierarchy

Access follows a three-tier administrative hierarchy plus the candidate and a platform analytics role. Each tenant (company) is fully isolated from every other.

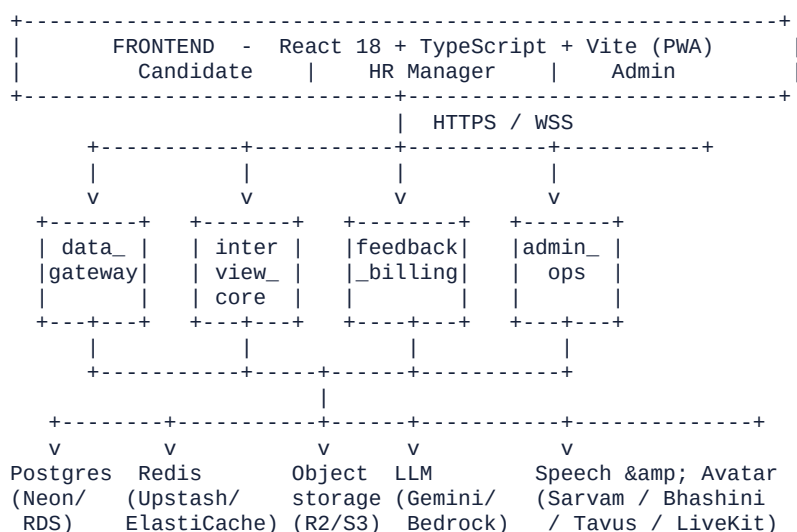
Role	Scope	Console	Key capabilities
platform_owner	Whole platform	/platform	Create/manage companies; create one super_admin per company; feature flags; DPDP audit
super_admin	Single company	/superadmin	Create & manage HR managers for its own company only
hr_manager	Single company	/hr	Resume screening, exams, interview invites, pipeline decisions
candidate	Self	Magic link	Take MCQ exams and AI voice interviews
admin	Platform analytics	/admin/*	Throughput, scores, system health dashboards

7.1 Provisioning Flow



8. Solution Architecture

Anterview is composed of **four backend microservices** behind a single React PWA, sharing managed datastores and pluggable AI / speech / avatar providers.



- **interview_core** - real-time WebRTC transport (LiveKit), the LangGraph conversation brain, the voice pipeline, and the per-turn interview loop.

- **data_gateway** - authentication (pluggable), user management, the Naipunyam SSO bridge, job sync, and DPDP consent.
- **feedback_billing** - AI scoring, scorecards, PDF generation, exam question generation, and the billing pipeline.
- **admin_ops** - admin dashboard, analytics, and operational APIs.

9. Technology Stack (Two-Tier)

The same codebase runs on a fast demo stack today and migrates to an AWS Mumbai production stack via configuration only - no code changes.

Layer	Demo (Tier 1)	Production (Tier 2)
LLM	Google Gemini	AWS Bedrock (Mumbai)
Speech (STT/TTS)	Sarvam + Bhashini	Bhashini + AI4Bharat
Avatar	Tavus / Simli via LiveKit	Three.js + Ready Player Me + Rhubarb
Database	Neon (Postgres + pgvector)	AWS RDS Mumbai (Postgres 16)
Cache	Upstash (Redis)	AWS ElastiCache (Redis 7)
Object storage	Cloudflare R2	AWS S3 Mumbai (SSE-KMS)
Email	Resend	AWS SES Mumbai
Hosting	Vercel + Railway	AWS EKS Mumbai (Multi-AZ)
Frontend	React 18 + TypeScript + Vite (PWA)	(same)
Backend	Python 3.12 + FastAPI + LangGraph	(same)
Auth	Pluggable (local/Google/Keycloak/Naipunyam)	(same)
Observability	Sentry	Prometheus + Grafana + Loki + OpenTelemetry

10. Functional Flow - Candidate AI Interview

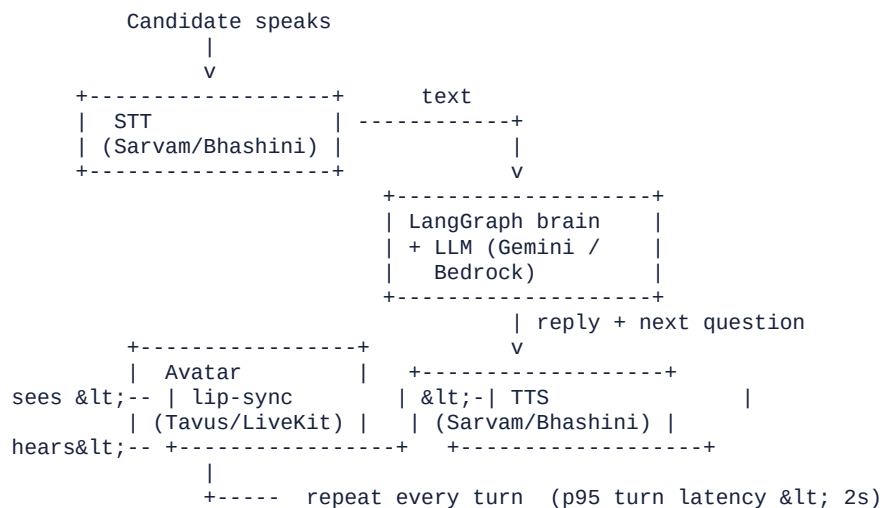


1. Candidate signs in (local / Google / SSO) or opens a secure interview link.
2. Candidate gives DPDP consent, then selects a job role and preferred language.
3. The avatar greets the candidate and starts a natural spoken conversation.

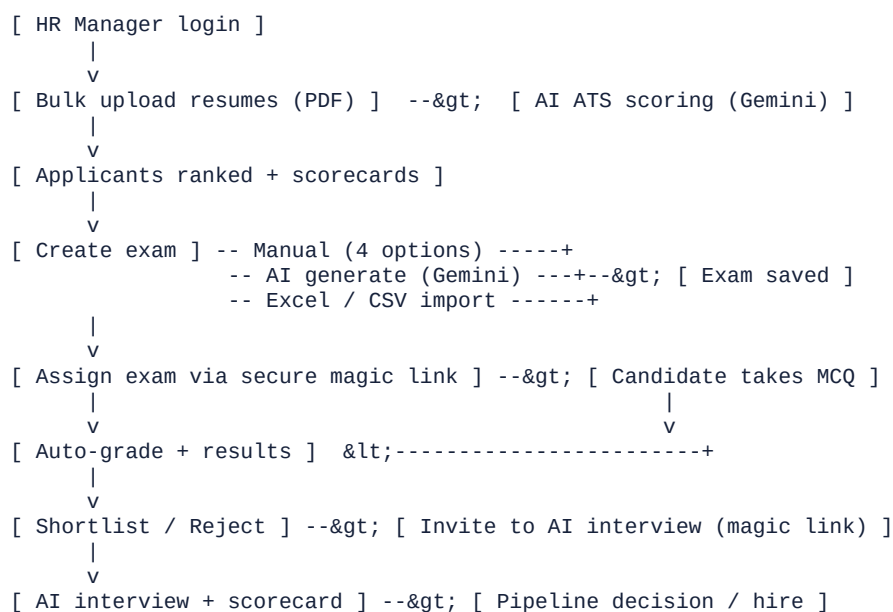
4. Each turn runs the real-time voice loop with adaptive follow-up questions.
5. After ~10 minutes (up to 10 questions) the session ends and is AI-scored.
6. A structured competency scorecard is shown on-screen and offered as a PDF.

11. Functional Flow - Real-Time Voice Pipeline

Every conversational turn flows through speech-to-text, the LangGraph brain, text-to-speech, and avatar lip-sync, engineered to keep **p95 turn latency under 2 seconds**.

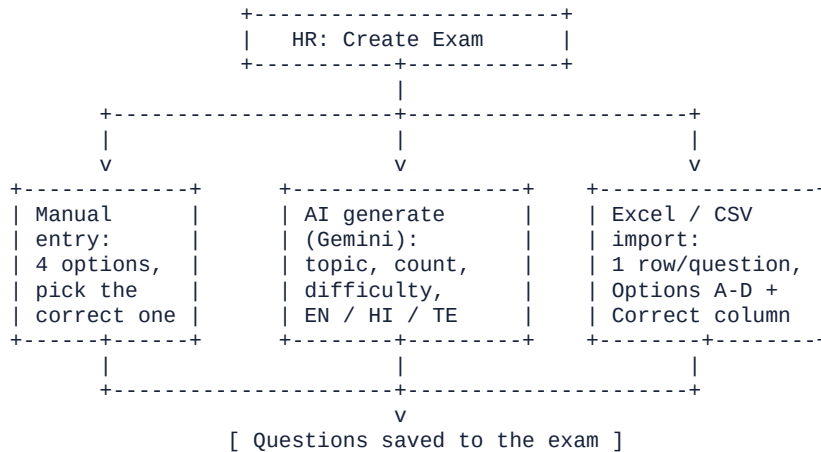


12. Functional Flow - HR Hiring Workflow (ATS)



1. HR uploads many resumes at once; names and emails are read from the PDFs.
2. Each resume is AI-scored against the role and ranked with a fit recommendation.
3. HR builds an exam using any of three methods (manual, AI, or Excel import).
4. The exam is assigned to candidates via secure, single-use magic links.
5. Candidates take the MCQ exam; it is auto-graded and results return to HR.
6. HR shortlists or rejects, then invites strong candidates to an AI interview.
7. Interview scorecards feed the hiring pipeline and final decision.

13. Exam Module - Three Authoring Methods



- **Manual** - type each question with four options and mark the correct answer.
- **AI generate** - Gemini drafts multiple-choice questions from a topic/role, count, difficulty, and language (EN/HI/TE); HR reviews before saving.
- **Excel / CSV import** - one row per question (Question, Option A-D, Correct, Points); a downloadable template is provided and invalid rows are reported.

14. Non-Functional Requirements

Requirement	Target / Commitment
Performance	p95 turn latency < 2 seconds
Availability	99.5% uptime, Multi-AZ design
Scalability	Up to 20 lakh users; 10-minute sessions
Avatars	6 avatars (3 male, 3 female) with selection
Languages	EN / HI / TE Day-1; 22 Indian languages roadmap
Data residency	India / Mumbai region for production
Cost	<= Rs. 12 variable cost per session (target Rs. 10)
Security	OWASP-aligned; pluggable auth; tenant isolation

15. Security, Privacy & Compliance (DPDP Act 2023)

- **Consent ledger** - every personal-data processing event is logged and auditable.
- **India data residency** - production data stays in the Mumbai region; no exceptions.
- **Right to erasure** - a retention policy purges expired sessions and cascaded data.
- **Tenant isolation** - every record is scoped by company; cross-tenant access returns 404.
- **Secure links** - exam/interview magic links are single-use, time-boxed, and stored only as HMAC hashes.
- **Secrets management** - no hardcoded secrets; all credentials are environment-injected.
- **PII discipline** - resume and transcript text is never written to application logs.

- **Pluggable auth + JWT** - short-lived access tokens, httpOnly refresh, CSRF protection.

16. Deliverables

- Running platform: 4 microservices + real-time interview worker + React PWA.
- Candidate interview experience with 6 avatars and EN/HI/TE voice.
- HR suite: resume screening, exam authoring (3 methods), magic-link assessments.
- Three-tier admin hierarchy and analytics dashboard.
- Scorecards (on-screen and PDF).
- Design documents (HLD, LLD, stack & economic model) and this sign-off document.
- Env-swappable configuration for the Tier-2 AWS Mumbai production path.

17. Assumptions, Dependencies & Constraints

- Third-party AI, speech, and avatar providers remain available within agreed quotas.
- Bhashini ULCA and AWS Bedrock approvals proceed for the production (Tier-2) path.
- Candidates have a modern browser, a microphone, and a basic internet connection.
- Day-1 language scope is EN / HI / TE; additional languages follow the roadmap.
- Production deployment is gated on a security-auditor sign-off.
- Per-session variable cost must remain $\leq$ Rs. 12 to preserve L1-bid viability.

18. Acceptance Criteria

The delivery is accepted when the following are demonstrated:

- A candidate completes an end-to-end AI voice interview and receives a PDF scorecard.
- HR uploads resumes, sees AI ATS scores, and authors an exam by all three methods.
- An assigned candidate takes an exam via magic link and HR sees auto-graded results.
- The admin hierarchy provisions companies, super admins, and HR managers correctly.
- The analytics dashboard reports live throughput, scores, and system health.
- Day-1 languages (EN/HI/TE) work for both voice and AI-generated content.
- DPDP consent is captured and tenant isolation is verified.

19. Project Phases

Phase	Description	Status
Phase 1+	Demo platform on managed cloud (4 services + worker + PWA)	Built & running
Hardening	Admin hierarchy, deletions, live stats, production hardening	Done
Tier-2 migration	AWS Mumbai (Bedrock, RDS, EKS) for residency + L1 bids	Planned (env-swappable)

20. Sign-Off & Acceptance

By signing below, the undersigned confirm that they have reviewed this document and agree to the scope, solution approach, non-functional requirements, and acceptance criteria for the Anterview AI Voice Interview Platform as described herein.

Role	Name	Designation	Signature	Date
Prepared by		Intants Engineering		
Reviewed by		Project / Engineering Lead		
Approved by		HR Head / Hiring Owner		
Approved by		Project Sponsor		

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